1 Dimension of a Subsapce

Theorem 1.1 (Dimension of a Subspace)

Let W be a subspace of a finite-dimensional vector space V. Then

(a) W is finite-dimensional.

(b) $\dim W \leq \dim V$.

(c) There exists a subspace U of V such that $V = W \otimes U$.

(d) $W = V$ if and only if $\dim W = \dim V$.

Proof:

(a) Since V is finite-dimensional, it has dimension m for some m. If W is not finite-dimensional, then we can choose an element in W w_0 and make a set $S = \{w_0\}$. If this is not a basis, we can add a element not in the span, w_1 to get an independent set. If we are in a non-finite dimensional space, this process can be done in such a way that it does not terminate (otherwise, we would obtain a finite basis and finite spanning set, making W finite dimensional). But, when we consider S as a subset of V, as soon as we obtain m independent vectors, we have a basis for V, implying that these span V, meaning they span W, a contradiction. So, we can never be in a non finite-dimensional space to begin with, so W is finite-dimensional.

(b) Take a basis for W, then consider it in V. Since this is independent in W over the field F, it is also independent in V (since we are working in the same subspace), we can extend this to a basis for V with the number of elements $\geq \dim W$, and since dimension is well defined, we therefore have $\dim V \geq \dim W$.

(c) As we did before, take a basis $\{w_i\}$ for W, and expand it to a basis for V by $\{w_1, ..., w_m, u_1, ..., u_n\}$, and let $W = \text{span}\{w_1\}$, $U = \text{span}\{u_1, ...,_n\}$. We claim $V = U \oplus W$. Since $v \in V$, we can write $v = \alpha_1 w_1 + ... + \alpha_m w_n + \beta_1 u_1 + ... + \beta_n u_n$ for some scalars $\{\alpha_i\}, \{\beta_i\}$. But, $w = \alpha_1 w_1 + ... + \alpha_m w_n \in W$, $u = \beta_1 u_1 + ... + \beta_n u_n \in U$, so $v = w + u$ for $w \in W$, $u \in U$. Now, we need to show the intersection is trivial. But, if we have $w = u$ for $w = \sum_{i=1}^{n} \alpha_i w_i$, $u = \sum_{i=1}^{n} \beta_i u_i$, then $w - u = 0$, so $\sum_{i=1}^{n+m} \gamma_i v_i = 0$ with $\gamma_i = \alpha_i, v_i = w_i$ for $1 \leq i \leq m$, and $\gamma_i = -\beta_i, w_i = u_i$ for $m + 1 \leq i \leq n + m$ is a linear combination of the basis vectors equal to zero, making all the γ_i equal to zero by independent, hence all the α_i, β_i are zero, i.e $w = u = 0$, showing triviality of the intersection.

(d) If $W = V$, the statement is trivial. Conversely, if $\dim W = \dim V$, consider the basis of W as an independent set in V. Either this already spans V, or we can append elements to make it span V. But, appending elements will give us a set of independent elements with more than $\dim V$ elements (since it already had $\dim W = \dim V$ elements), so the only possibility is that it already spanned V, so $V \subset \text{span}\{w_1, ..., w_n\} = W \subset V$, making $W = V$.

Example 1.2

Let W be the subspace of $\mathbf{R}^5$ defined by $W = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{R}^5, x_1 = 3x_2 \text{ and } x_3 = 7x^4\}$.

To find a basis of U, first note that:

$$(3x_2, x_2, 7x_4, x_4, 5) = x_2(3, 1, 0, 0, 0) + x_4(0, 0, 7, 1, 0) + x_5(0, 0, 0, 0, 1)$$

shows that W is spanned by $\{(3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1)\}$. Since:

$$(0, 0, 0, 0, 0) = \alpha_1(3, 1, 0, 0, 0) + \alpha_2(0, 0, 7, 1, 0) + \alpha_3(0, 0, 0, 0, 1) = (3\alpha_1, \alpha_1, 7\alpha_2, \alpha_2, \alpha_3)$$

implies $\alpha_i = 0$ for all i, this set is independent and therefore a basis of W.

By the plus-minus lemma, we can extend this to a basis of $\mathbf{R}^5$. We can add $(1,0,0,0,0)$ since it is not in the span of the prior vectors, and it follows this is independent. Then, once we add this, $(0,0,1,0,0)$ is not in the span of these, so we can add this. Then, we have five factors independent (by plus/minus lemma), so this is a basis of $\mathbf{R}^5$, and furthermore R^5 decomposes as $W \oplus U$ with $U = \text{span}\{(1,0,0,0,0),(0,0,1,0,0)\}$.



Exercise 1.3

Let $W = \{p \in \mathcal{P}_4(F) : p(6) = 0\}$.

(a) Find a basis of W.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(F)$.

(c) Find a subspace U of $\mathcal{P}_4(F)$ such that $\mathcal{P}_4(F) = W \oplus U$.



Solution 1

1. These are exactly the polynomials with degree at most 4 with 6 as a root, that is, those polynomials that factor as $p(x) = (x-6)q(x)$ with $q(x)$ a polynomial in $\mathcal{P}_3(x)$. A basis for these will arise from finding a basis for the $q(x)$ and multiplying these by $x-6$ (to give a subspace in x^6). These are $(x-6) = (x-6)\cdot 1, x^2-6x = (x-6)\cdot x, x^3-6x^2 = (x-6)\cdot x^2, x^4-6x^3 = (x-6)\cdot x^3$. This clearly spans the subspace, and is independent since if we have a sum of these $(x-6)\cdot 1, x, x^2, x^3$ equal to zero, then it factors as $(x-6)\left(\sum_{i=0}^{3}\alpha_i x^i\right) = 0$, and this is zero if and only if $\left(\sum_{i=0}^{3}\alpha_i x^i\right) = 0$, which (since this is a basis for $\mathcal{P}_3(x)$) forces all the $\alpha_i = 0$.

2. Since this is dimension four, simply adding an vector not in the span of these will give a linearly independent set with 5 elements in V by the plus-minus lemma, which must be a basis for V since this subspace is of $\dim = 5$ and $\dim V = 5$. Indeed, $v = 1$ is not in this, so this extends to a basis $\{1,(x-6),(x-6)x,(x-6)x^2,(x-6)x^3\}$ for V.

3. Notice this is exactly the same construction as in part c of Theorem 1.1, and indeed this is a direct sum by the same argument as in there.



Exercise 1.4

Suppose W and U are subspaces of V such that $V = W \otimes U$. Suppose also that $\{w_1,...,w_m\}$ is a basis of W and $\{u_1,...,u_n\}$ is a basis of U. Prove that $\{w_1,...,w_m,u_1,...,u_n\}$ is a basis of V.



Proof:

We need to show span and independence. Since $V = W \otimes U$, we can write $v = w + u$ for $w \in W$, $u \in U$. But, all $w \in W$ are of the form $\sum_{i=1}^{m}\alpha_i v^i$, and the $u \in U$ are of the form $\sum_{i=1}^{n}\beta_i u^i$, so $v = \sum_{i=1}^{n+m}\gamma_i u_i$ with $\gamma_i = \alpha_i, v_i = w_i$ for $1 \le i \le m$, and $\gamma_i = \beta_i, w_i = u_i$ for $m+1 \le i \le n+m$, showing that the $\{w_1,...,w_m,u_1,...,u_n\}$ span V.

Now, we show independence. Consider any sum $\sum_{i=1}^{n+m}\gamma_i u_i = 0$ with $\gamma_i = \alpha_i, v_i = w_i$ for $1 \le i \le m$, and $\gamma_i = -\beta_i, w_i = u_i$ for $m+1 \le i \le n+m$ (this here defines the α_i, β_i). Then, this equation can be rearranged as $\sum_{i=1}^{m}\alpha_i w_i = \sum_{i=1}^{n}\beta_i u_i$, so $w = u$, but since this is a direct sum, the intersection is trivial, so the α_i, β_i are zero, so the γ_i are all zero, showing independence.



Example 1.5

We will prove that the subspaces of $\mathbf{R}^2$ are precisely the trivial subspace, the lines in $\mathbf{R}^2$ through the origin, and $\mathbf{R}^2$ itself. Since $\dim \mathbf{R}^2 = 2$, by Theorem 1.1, any subspace must have dimension $0, 1, 2$.

- The only 0-dimensional subspace is the trivial subspace.
- If W is a 1-d subspace, we have the span of a non-zero vector (x_1, x_2), that is, elements of the form $(v_1, v_2) = (\alpha x_1, \alpha x_2)$, but this is exactly the parametric equation of a line through $(0, 0)$ and (x_1, x_2).
- If dimension $2 = \dim \mathbf{R}^2$, by Theorem 1.1, this is the whole space.

Exercise 1.6

Prove that the subspaces of $\mathbf{R}^3$ are precisely $\{0\}$, the lines through the origin, all planes in $\mathbf{R}^3$ through the origin, and $\mathbf{R}^3$.

Proof:

Subspaces can only be of dim $\leq \dim \mathbf{R}^3 = 3$

- If 0-dim, then $\{0\}$.
- If 1-dim, then we have exactly the span of one vector, so $(v_1, v_2, v_3) = (\alpha x_1, \alpha x_2, \alpha x_3)$, i.e a parametrization of the line through $(0, 0, 0)$ and (x_1, x_2, x_3).
- The 2-dimensional subspaces are spans of two vectors, that is, elements of the form $(v_1, v_2, v_3) = \alpha(x_1, x_2, x_3) + \beta(y_1, y_2, y_3)$, where (x_1, x_2, x_3) are not multiples of each other. But, this is of course a plane in $\mathbf{R}^2$ through the origin (if you want a parametrization of the plane, take $\vec{x} \times \vec{y}$ to get a normal vector, this is guaranteed to work since x, y are not multiples of each other).
- 3-d subspace is exactly $\mathbf{R}^3$.



2 Dimension of a Sum

The next theorem is analogous to the formula

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|$$

for the number of elements in the union of finite sets.



Theorem 2.1 (Dimension of a Sum)

If V_1 and V_2 are subspaces of a finite-dimensional vector space, then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

Exercise 2.2

Fill in the steps in the proof of theorem 2.1 below.

Proof:

Let $\{v_1, ..., v_m\}$ be a basis of $V_1 \cap V_2$. Then $\dim(V_1 \cap V_2) = m$. Since $\{v_1, ..., v_m\}$ is a basis of $V_! \cap V_2 \subset V_1$, it is linearly independent in V_1. Hence this list can be extended to a basis $\{v_1, ..., v_m, u_1, ..., u_j\}$ of V_1 by plus/minus, $\dim V_1 = m + j$. Similarly, we can extend $\{v_1, ..., v_m\}$ to a basis $\{v_1, ..., v_m, w_1, ..., w_k\}$ of V_2, so $\dim V_2 = m + k$.

We will show that

$$\{v_1, ..., v_m, u_1, ..., u_j, w_1, ..., w_k\}$$

is a basis of $V_1 + V_2$.

(a) Explain why this will complete the proof.

(b) Show that $V_1 + V_2 = \text{span}\{v_1, ..., v_m, u_1, ..., u_j, w_1, ..., w_k\}$.

(c) Complete the proof by showing that $\{v_1, ..., v_m, u_1, ..., u_j, w_1, ..., w_k\}$ is linearly independent.

Proof:

a If this is a basis, then $\dim V_1 + V_2 = m + j + k = (m + j) + (m + k) - m = \dim V_1 + \dim V_2 - \dim V_1 \cap V_2$.

b We know that $v \in V_1 + V_2$ is of the form $v = v^1 + v^2$ with $v^1 \in V_1$, $v^2 \in V_2$. But, since the $\{v_1, ..., v_m, u_1 u_j\}$ are a basis of V_1, we can write v^1 as a linear combination of v_i, u_i. Similarly, we can write v^2 as a linear combination of v_i, w_j, and then adding v^1 and v^2 (and combining the coefficients of the v_i since they appear twice) shows v is a linear combination of v_i, w_i, u_i, and we are done.

c Consider any linear combination of these equal to zero, by:

$$\sum_{i=1}^{m} \alpha_i v_i + \sum_{i=1}^{j} \beta_i u_i + \sum_{i=1}^{k} \gamma_i \beta_i = 0.$$

Rearranging, we have:

$$\sum_{i=1}^{m} \alpha_i v_i + \sum_{i=1}^{j} \beta_i u_i = -\sum_{i=1}^{k} \gamma_i \beta_i.$$

We claim that the right-hand term $-\sum_{i=1}^{k} \gamma_i \beta_i$ is not an element of V_1. We know that $V_2 = (V_2 \cap V_1) \oplus U$ with U the span of the $\{u_i\}$ (by Theorem 1.1, since this was the construction of the set), we know that any element in U (including the right-hand term above) cannot be nontrivial in $V_2 \cap V_1$. But, any such element is in V_2 since $U \subset V_2$, so to satisfy this condition, we must have that this element is not in V_1 (unless it is trivial).

Applying this to this term, we know the RHS is not nontrivial and in V_1, but the LHS is a linear combination of elements in V_1 and therefore is in V_2. So, we must have that the RHS is trivial, meaning all the γ_i are zero. It follows from independence of the v_1, w_i that the α_i, β_i are zero.

Corollary 2.3

If V_1, V_2 are subspaces of a vector space such that $V = V_1 \oplus V_2$, then

$$\dim V = \dim V_1 + \dim V_2.$$



Proof:

If this is a direct sum, we see $V_1 + V_2 = \{0\}$, so applying the prior theorem gives this directly.





Exercise 2.4

Suppose U and W are both five-dimensional subspaces of $\mathbf{R}^9$. Prove that $U \cap W \neq \{0\}$.



Proof:

Consider the direct sum $U + W$, which is itself a subspace of $\mathbf{R}^9$, so $\dim(U + W) \leq 9$. If the intersection is trivial, we have;

$$\begin{aligned} \dim(U + W) &= \dim(U) + \dim(W) - \dim(U \cap W) \\ &= 5 + 5 - \dim(\{0\}) \\ &= 10, \end{aligned}$$

but $10 \not\leq 9$, a contradiction. So, this cannot happen.

